Vectors in 3 Dimensions

Exam code: 9MA0

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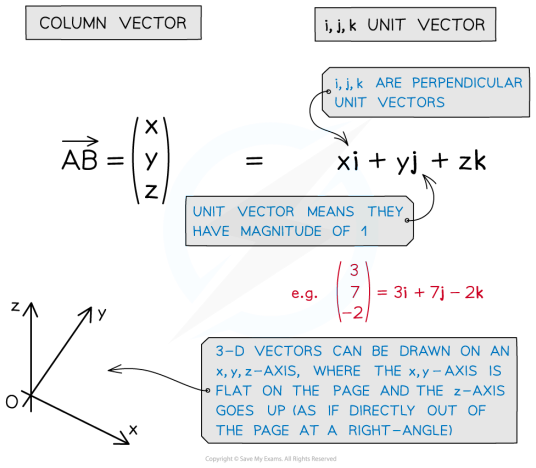
Vectors in 3 dimensions

What is a 3D vector?

- Vectors represent a movement of a certain **magnitude** (size) in a given **direction**
  You should have already come across (2D) **vectors** at AS (see Basic Vectors)
- 3D vectors describe the **position of a point** in a 3D space in relation to the origin
- They can be represented in different ways such as a **column vector** or in **i, j, k unit vector form**



Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- Like 2D vectors we can find the **magnitude** using Pythagoras' theorem (see Magnitude Direction)

MAGNITUDE

$$|\mathbf{a}| = |x\mathbf{i} + y\mathbf{j} + z\mathbf{k}| = \left|\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right| = \sqrt{x^2 + y^2 + z^2}$$

REMEMBER THERE ARE LOTS OF DIFFERENT WAYS TO REPRESENT THE SAME VECTOR

$$|\mathbf{a}| = |\overrightarrow{AB}| = \left|\begin{pmatrix} 3 \\ 7 \\ -2 \end{pmatrix}\right| = |3\mathbf{i} + 7\mathbf{j} - 2\mathbf{k}|$$

A VECTOR'S MAGNITUDE IS SOMETIMES REFERRED TO AS ITS MODULUS

$$|\overrightarrow{AB}| = \sqrt{3^2 + 7^2 + 2^2} = \sqrt{62} = 7.874\ldots = 7.9 \text{ (1 dp)}$$

YOU CAN IGNORE MINUS SIGN

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- For 3D **position vectors** we can find the distance between two points
- By using the respective co-ordinates we can calculate the magnitude of the vector between them:

DISTANCE BETWEEN TWO POINTS FROM POSITION VECTORS

(x_1, y_1, z_1) AND $(x_2, y_2, z_2) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$

WHERE x, y, z ARE FROM POSITION VECTORS $x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$

e.g. FIND $|\overrightarrow{AB}|$ FOR POSITION VECTORS A AND B, $5\mathbf{i} + 3\mathbf{j} - 7\mathbf{k}$ AND $3\mathbf{i} - 6\mathbf{j} + 5\mathbf{k}$ RESPECTIVELY

A HAS CO-ORDINATES $(5, 3, -7)$

B HAS CO-ORDINATES $(3, -6, 5)$

$$|\overrightarrow{AB}| = \sqrt{(5-3)^2 + (3 - -6)^2 + (-7-5)^2} = \sqrt{4 + 81 + 144}$$

$$= \sqrt{229} = 15.1 \text{ UNITS (1 dp)}$$

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3D vector addition, scalars, parallel vectors and unit vectors

- 3D vectors work in the same way as 2D vectors, just in three dimensions rather than two
- **Vector addition** and subtraction and scalar multiplication can be carried out in exactly the same way, this time involving **i, j** and **k** or x, y and z
- 3D vectors are also **parallel** if one is a **multiple** of the other

$\mathbf{a} = \begin{pmatrix} 6 \\ -4 \end{pmatrix}$ $\mathbf{b} = \begin{pmatrix} 7 \\ 4 \end{pmatrix}$

$\mathbf{b} = 8\mathbf{i} - 6\mathbf{j} + 4\mathbf{k}$

$$2\mathbf{a} - 3\mathbf{b} = 2 \times \begin{pmatrix} 2 \\ 6 \\ -4 \end{pmatrix} - 3 \times \begin{pmatrix} -3 \\ 7 \\ 4 \end{pmatrix}$$

$$= \begin{pmatrix} 4 \\ 12 \\ -8 \end{pmatrix} - \begin{pmatrix} -9 \\ 21 \\ 12 \end{pmatrix} = \begin{pmatrix} 13 \\ -9 \\ -20 \end{pmatrix}$$

$$2\mathbf{a} + \frac{1}{2}\mathbf{b} = 2(-3\mathbf{i} + 9\mathbf{j} + 2\mathbf{k}) + \frac{1}{2}(8\mathbf{i} - 6\mathbf{j} + 4\mathbf{k})$$

$$= (-6\mathbf{i} + 18\mathbf{j} + 4\mathbf{k}) + (4\mathbf{i} - 3\mathbf{j} + 2\mathbf{k})$$

$$= -2\mathbf{i} + 15\mathbf{j} + 6\mathbf{k}$$

3-D VECTORS

PARALLEL VECTORS

$\mathbf{a} = \begin{pmatrix} 9 \\ -6 \\ 3 \end{pmatrix}$ $\mathbf{b} = \begin{pmatrix} 6 \\ -4 \\ 2 \end{pmatrix}$

$\mathbf{a}$ IS PARALLEL TO $\mathbf{b}$ AS

$\mathbf{b} = \frac{2}{3}\mathbf{a}$ $\frac{2}{3}\begin{pmatrix} 9 \\ -6 \\ 3 \end{pmatrix} = \begin{pmatrix} 6 \\ -4 \\ 2 \end{pmatrix}$

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- **Unit vectors** in 3D are found in exactly the same way as in 2D

TO FIND THE **UNIT VECTOR** IN THE SAME DIRECTION AS A GIVEN VECTOR, SIMPLY DIVIDE THE VECTOR BY ITS MAGNITUDE

- THE UNIT VECTOR IN THE SAME DIRECTION AS VECTOR $\overrightarrow{AB}$ IS $\frac{\overrightarrow{AB}}{|\overrightarrow{AB}|}$
- THE UNIT VECTOR IN THE SAME DIRECTION AS VECTOR $\mathbf{a}$ IS $\frac{\mathbf{a}}{|\mathbf{a}|}$

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e.g. FIND THE UNIT VECTOR IN THE SAME DIRECTION AS THE VECTOR $\mathbf{a} = 3\mathbf{i} - 4\mathbf{j} + 5\mathbf{k}$

$$|\mathbf{a}| = \sqrt{3^2 + (-4)^2 + 5^2} = 5\sqrt{2}$$

$$\text{UNIT VECTOR} = \frac{\mathbf{a}}{|\mathbf{a}|} = \frac{3\mathbf{i} - 4\mathbf{j} + 5\mathbf{k}}{5\sqrt{2}} = \frac{1}{5\sqrt{2}}(3\mathbf{i} - 4\mathbf{j} + 5\mathbf{k})$$

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